

* If a total of 33 MHz of bandwidth is allocated to a particular TDD cellular telephone system, which uses two 25 kHz simplex channels to provide half duplex voice and control channels, compute the number of channels available per cell if a system uses (a) 4-cell reuse, (b) 7-cell reuse, (c) 12-cell reuse. If 1 MHz of the allocation spectrum is dedicated to control channels, determine an equitable distribution of control channels and voice channels in each cell for each of the three systems.

Ans: Total available bandwidth = 33 MHz

Simplex channel bandwidth = 25 KHz

Spectrum reserved for control channel = 1 MHz

Each duplex channel requires 2 simplex channels. So, the bandwidth per duplex channel = $2 \times 25 \text{ KHz} = 50 \text{ KHz}$ per duplex channel.

Available bandwidth for duplex voice channels, $33 - 1 = 32 \text{ MHz}$

Total no. of duplex voice channels available is

$$\frac{32,000 \text{ KHz}}{50 \text{ KHz}} = 640$$

Total no. of ~~control~~ half-duplex control channels available,

$$\frac{1000 \text{ KHz}}{25 \text{ KHz}} = 40$$

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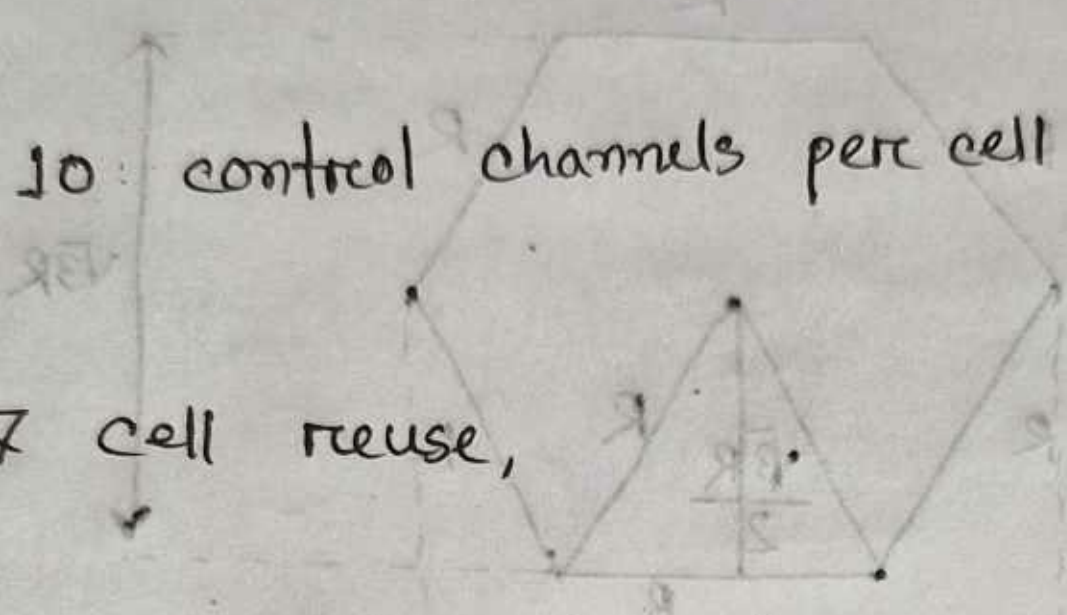
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a) For $N=4$ cell reuse,

$$\frac{640}{4} = 160 \text{ voice channels per cell}$$

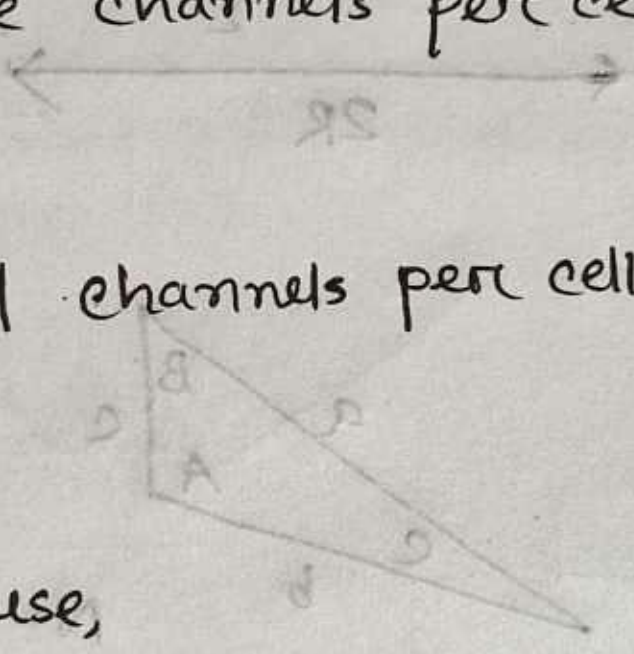
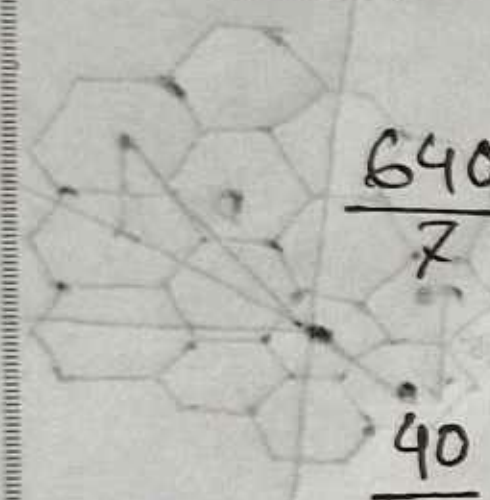
$$\frac{40}{4} = 10 \text{ control channels per cell}$$



b) For $N=7$ cell reuse,

$$\frac{640}{7} \approx 91 \text{ voice channels per cell}$$

$$\frac{40}{7} \approx 6 \text{ control channels per cell}$$



c) For $N=12$ cell reuse,

$$\frac{640}{12} \approx 53 \text{ voice channels per cell}$$

$$\frac{40}{12} \approx 4 \text{ control channels per cell}$$

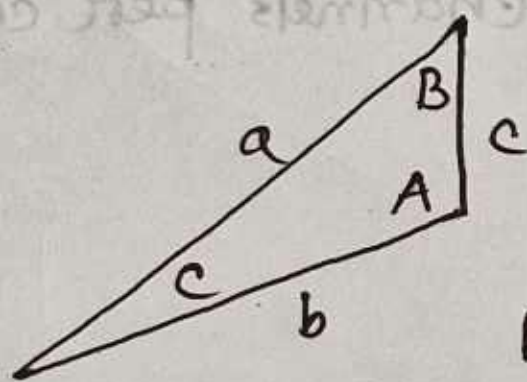
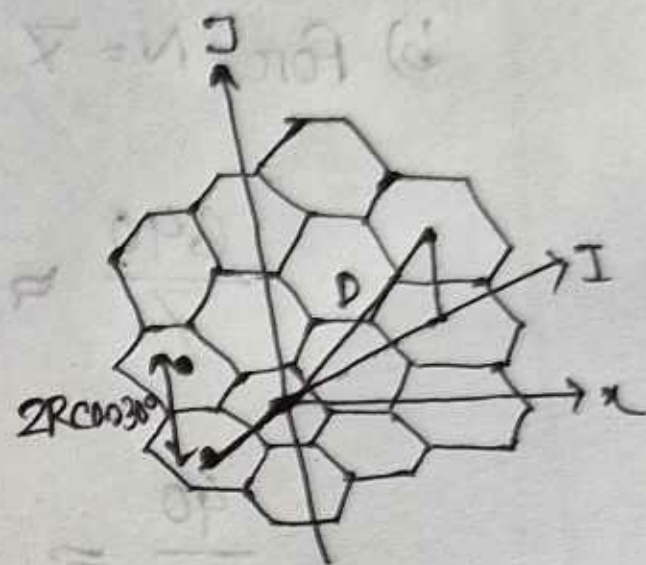
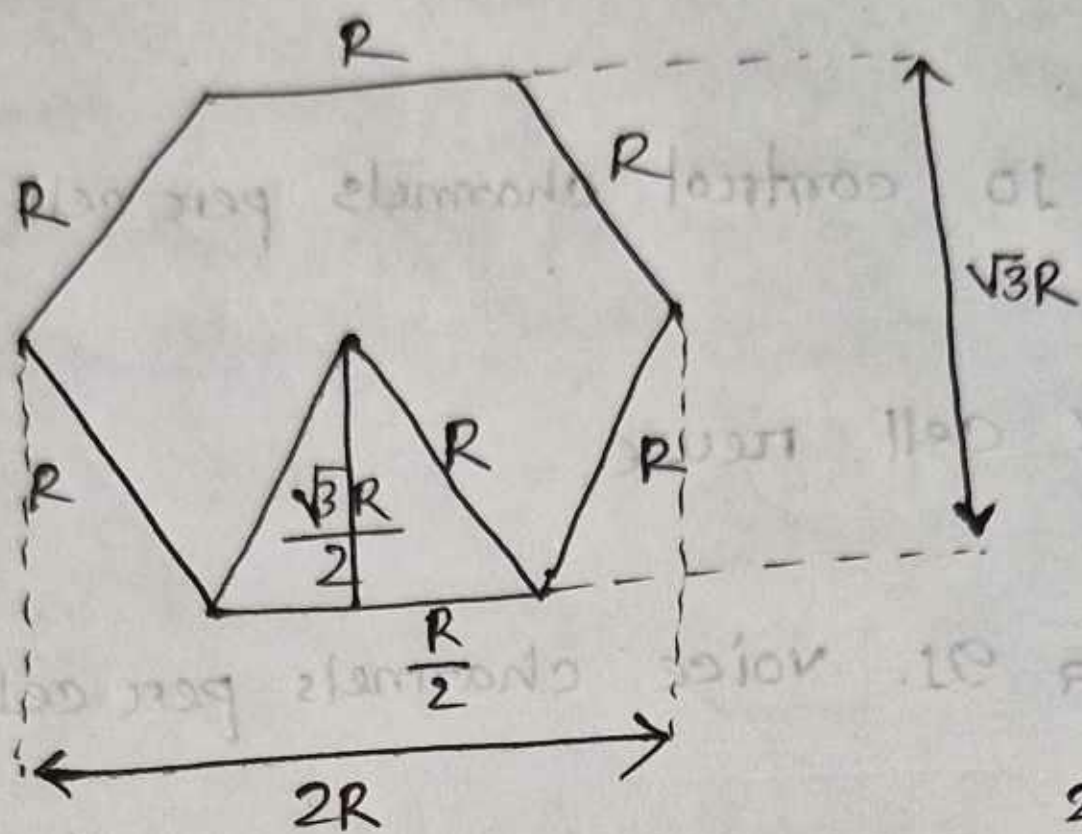
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* Prove that for a hexagonal geometry, the co-channel reuse ratio is given by $Q = \sqrt{3}N$, where $N = i^2 + j^2 + ij$

Ans:



$$N = i^2 + j^2 + ij \quad \text{--- (1)}$$

Cosine rule, $a^2 = b^2 + c^2 - 2bc \cos(A)$

$$d = 2R \cos 30^\circ = 2R \frac{\sqrt{3}}{2} = \sqrt{3}R \quad \text{--- (II)}$$

$$D^2 = (i \times d)^2 + (j \times d)^2 - 2(i \times d)(j \times d) \cos 120^\circ$$

$$= i^2 d^2 + j^2 d^2 - 2ij d^2 \cos 120^\circ$$

$$= d^2 (i^2 + j^2 - 2ij (-\frac{1}{2}))$$

$$= d^2 (i^2 + j^2 + 2ij)$$

$$= (\sqrt{3}R)^2 N$$

$$= 3R^2 N$$

$$\therefore D = \sqrt{3N} R$$

$$Q = \frac{D}{R} = \frac{\sqrt{3N} \cdot R}{R} = \sqrt{3N} \quad (\text{Proved})$$

Chapter 04 (GSM)

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* GSM functional architecture

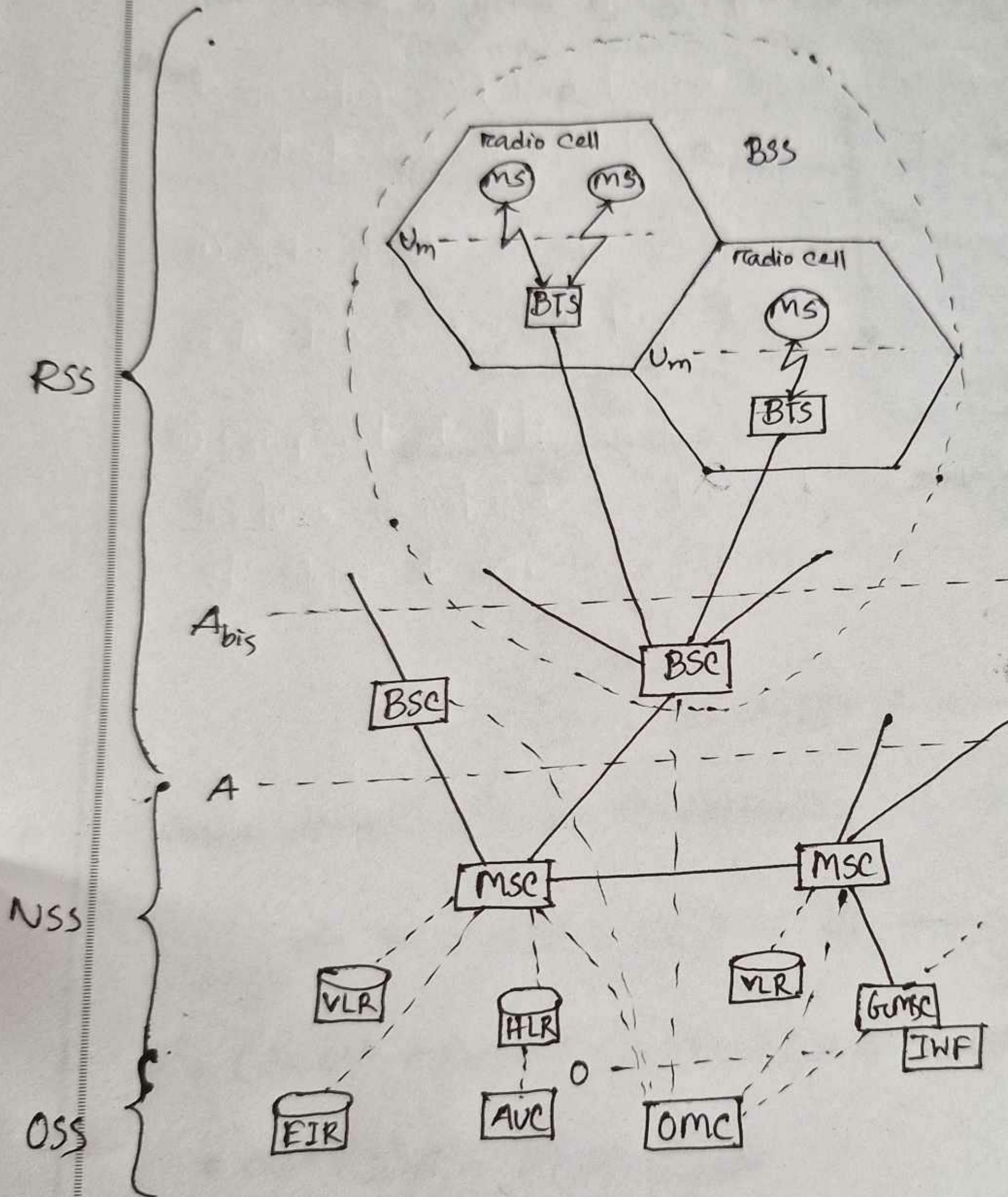


Fig: System architecture

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Components: A GSM system consist of 3 subsystems.

1. Radio Subsystem: comprises all radio specific entities.

i) mobile station (MS) - The user's mobile device, consisting of the mobile equipment and Subscriber Identity Module (SIM) card.

ii) Base Transceiver Station (BTS) - contains radio equipments like antennas, signal processing, etc. Connected to MS via UM interface

iii) ~~BSS~~ Base station subsystem (BSS) - performs all functions necessary to maintain radio connections to an MS. It contains several BTSs and BSCs.

iv) Base station controller (BSC) - manages the BTSs, handles handover, performs paging of the MS. It is connected to BTS via Abis interface

2. Network and Switching Subsystem

i) mobile services switching center (MSC) - set up connection other MSCs and to BSCs via A interface. Manages the BSC

ii) Home location register (HLR) - important database that stores all user-relevant data information.

iii) Visitor location register (VLR) - a dynamic database works in coordination of the HLR. Stores MS users currently in location

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3. Operation Subsystem: contains necessary functions for network operation and maintenance.

i) Operation and Maintenance Center (OMC) - monitors and controls all other network entities via the O interface

ii) Authentication center (AUC) - protects user identity and data transmission. contains the algorithm for authentication.

iii) Equipment identity register (EIR) - stores all device identifications registered on this network. It also has blacklist of stolen devices.